

WASP-104b

R-2302

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_221203 · created 2026-08-17T23:10:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459916.9321 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.044
Transit depth (Rp/Rs)^2	2.19 +/- 1.31 [%]
Orbital Inclination (inc)	87.04 +/- 2.93
Airmass coefficient 1 (a1)	12.83 +/- 11.96
Airmass coefficient 2 (a2)	-1.34 +/- 0.57
Scatter in the residuals of the lightcurve fit is	136.79 %
Best Comparison Star	#1 - [216, 287]
Optimal Aperture	4.19
Optimal Annulus	5.54
Transit Duration (day)	0.089 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.044 vs 0.1214 (deviation 0.54σ)
Duration fit vs published	0.089 d vs 0.0742 d (deviation 0.83σ)
Mid-time O–C	12.6 min
Depth SNR	3.0
Residual scatter	136.79 %

Light curve

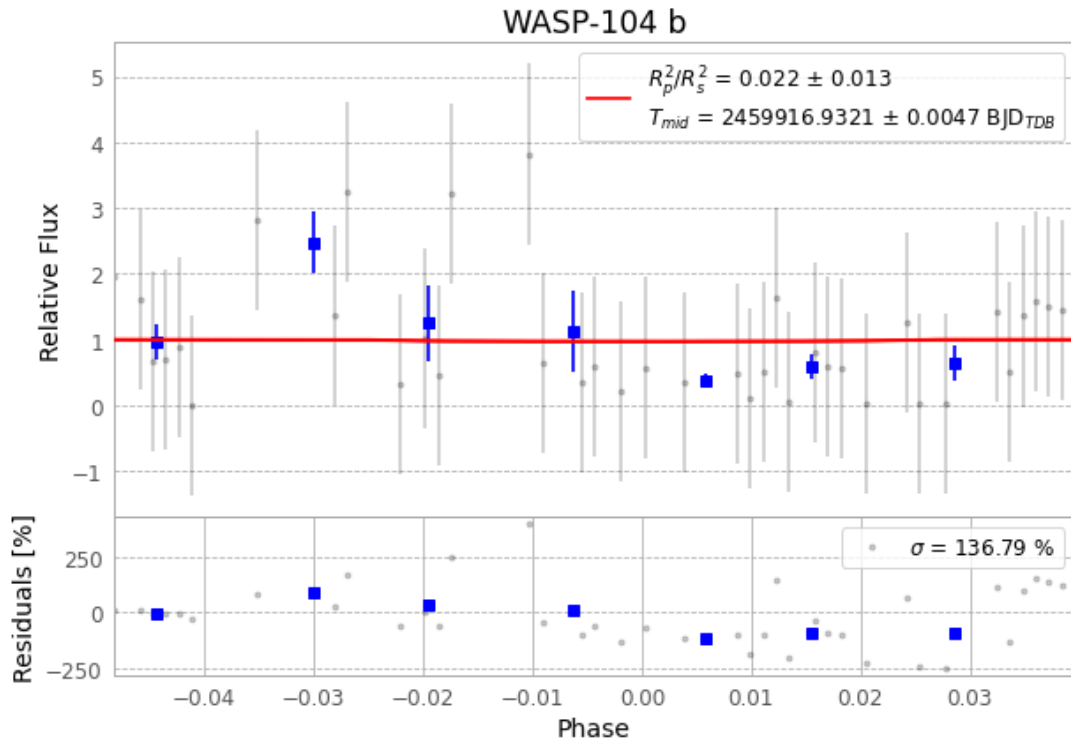


Figure 1 — FinalLightCurve_WASP-104 b_2022-12-03.png (SHA-256 cc53393ddde7408c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [448, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8a706e6a606bb1ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

75 FITS frames (49 MB), WASP-104221203081815.FITS → WASP-104221203120015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-221203.log (b50023fcd5d9...), FinalLightCurve_WASP-104 b_2022-12-03.png (cc53393ddde7...), FinalLightCurve_WASP-104 b_2022-12-03.csv (a344537c8a3e...)

Compute resources

Wall time 194.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:10Z.